

Dreaming of Butterflies

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Figure 1: Seated mixed reality installation with ink-style virtual spatial elements.

ABSTRACT

Dreaming of Butterflies is a mixed reality installation that examines how understanding can emerge through perception rather than instruction. A butterfly guides visitors from a quiet tea room through three interconnected scenes and back again, shaping attention through subtle pacing and minimal gestures such as gentle reach and slow turn. The work explores how perceptual orientation shifts when meaning arises through embodied sensing instead of explicit direction. The installation runs on Apple Vision Pro with a real-time ink rendering pipeline and is presented within a 3x3 meter seated tea room environment. It investigates the relationship

between guidance, attention, and embodied experience in mixed reality.

Index Terms: Mixed Reality, Embodied Interaction, Spatial Narrative, Ink Rendering, Apple Vision Pro, Zhuangzi.

1 DESCRIPTION OF ARTWORK

Dreaming of Butterflies is a mixed reality installation that invites visitors to learn through perception rather than explanation. The work draws inspiration from Zhuangzi's "Butterfly Dream" not as a story to be retold, but as a mode of understanding that arises through attention, quietness, and embodied encounter. The butterfly serves as a gentle guide that shapes the rhythm of attention and the pacing of perception. Instead of providing instruction, it creates conditions in which meaning has time to gradually surface through the body.

The audience journey begins in a calm tea room anchored to the actual site. A butterfly settles on a teacup placed on a low table. A gentle reach opens a cyclical passage through three linked scenes and then returns the visitor to the tea room. The first cultivates effortless action. Flowers respond to gaze and cups fill to light pinches so that acting without strain is felt through the body. The

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second aligns drifting platforms into a path when the point of view shifts. Coherence is discovered by moving and by trying alternative perspectives, which echoes relational truth in Daoist writing. The third turns floating text into fish and then into a soaring Kunpeng, a mythic giant bird, so that identity is sensed as a becoming rather than a fixed category. The return to the tea room offers no resolution. It appears unchanged, yet the visitor's sensing of it has shifted. What is learned comes from attending rather than being told, and this shift in perception prepares the ground for how the work is experienced.

The interaction grammar is intentionally small and natural, consisting of gentle reach, open hand, and slow turn. No explicit instructions are required. Sound, timing, and visual transitions follow a restrained logic that supports a contemplative tempo. The butterfly's proximity, hovering, and moments of shared gaze shape the rhythm of the experience rather than defining tasks, encouraging visitors to remain attentive rather than goal-oriented. In this way, the interaction structure itself sustains the perceptual openness introduced by the narrative journey, preserving agency by allowing discovery to unfold at each visitor's own pace.

Technically, the installation runs on Apple Vision Pro with inside-out tracking and native hand and eye input. Unity XR Interaction Toolkit and PolySpatial support a single interaction grammar that remains consistent across mixed and virtual space. A real-time ink rendering pipeline authored in Shader Graph shapes tone and edge, with noise-modulated alpha suggesting the hollows of feibai brushwork and layered diffusion allowing virtual forms to settle naturally into the live camera feed. All content runs locally on the headset without external sensing equipment, ensuring that the visual and interaction tone remains continuous across mixed and virtual space.

The installation occupies a 3x3 meter space arranged as a quiet tea room. A low table and a single floor cushion sit at the center, with clear walkways surrounding them. The experience begins seated, and all interactions occur within a one-meter reach, allowing the work to be completed comfortably without standing or moving around the room. The system runs entirely on Apple Vision Pro and requires no external tracking equipment. Soft, even ambient lighting supports hand tracking and visual comfort. Sound is conveyed directly through the headset, maintaining an intimate acoustic field that does not disturb adjacent works. One facilitator can host approximately eight to ten visitors per hour, depending on dwell time. The installation operates offline and stores no biometric or identifying data, supporting reliable and low-overhead exhibition in a range of public gallery contexts.

The contribution of this work lies in its integration of guidance, embodiment, and visual language. First, the interaction structure supports reflective engagement rather than task completion, allowing meaning to arise through perception. Second, the work offers a spatial and embodied interpretation of classical philosophy. Third, the lightweight real-time ink aesthetic makes traditional visual sensibility transferable to contemporary mixed reality exhibition. The piece creates a space in which memory, presence, and imagination coexist without closure, allowing the room to continue dreaming alongside the visitor.

2 ARTIST INFORMATION

Yifan Ding is a graduate student in Design at the School of Animation and Digital Arts, Communication University of China. Her research focuses on extended reality (XR), including virtual reality (VR) and mixed reality (MR), with an emphasis on embodied interaction and spatial narrative. She explores how minimal gesture vocabularies and contemplative temporal pacing shape audience perception in immersive environments.

Xin Lyu is a professor at the School of Animation and Digital Arts, CUC, and a Ph.D. supervisor in digital media art. His

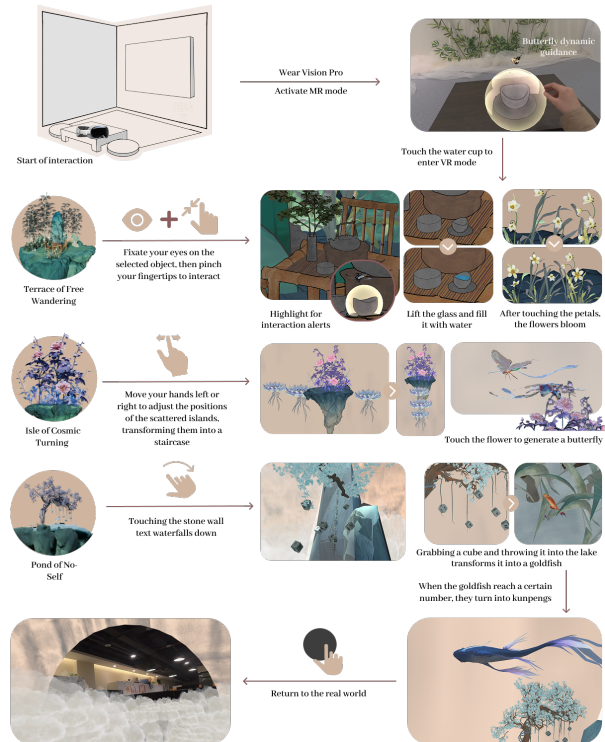


Figure 2: Interaction flow of *Dreaming of Butterflies* in an XR tea-room installation.

work spans internet product design, digital media art, and network communication, with long-term practice in curriculum building and interdisciplinary collaboration.

Wanxiang Ma is a visual designer and new media artist whose research focuses on cross-media design practices centered on Mixed Reality (MR). She explores the relationships between technology, perception, and social issues, and is committed to investigating how contemporary media shape human experience and cognition through immersive interaction, spatial storytelling, and data-driven visual systems. She is currently an undergraduate student in New Media Art at the Communication University of China, with a practice-based background that integrates visual design, interactive technologies, and conceptual research. Her work spans XR experiences, interactive installations, and experimental visual systems.

Sibo Yang is a lecturer at the School of Animation and Digital Arts, CUC. His research and teaching focus on interactive media and spatial computing, with experience in mixed reality prototyping and exhibition-oriented installations.

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